QISKIT FALL FEST 2026 — SRM UNIVERSITY-AP | TRACK: OPTIMIZATION

Quantum Optimization of Network Partitioning Using QAOA

Problem PS-O1 • Hardware Co-Design Paradigm • Team VARANASI

EXECUTIVE HYPOTHESIS & SCIENTIFIC VERIFICATION

NISQ algorithms waste critical coherence on routing overhead. By co-designing Processor D to natively embed the PS-O1 interaction graph, we eliminate 100% of routing SWAPs, reducing physical CX gates by 14.3% and circuit depth by 21.6%, while delivering an optimal solution sampling probability of 12.50% (a 4.0x enhancement over uniform random guessing).

0 SWAPs

-100% Routing Overhead

18 CX Gates

-14.29% vs Generic B (21)

Depth 29

-21.62% vs Generic B (37)

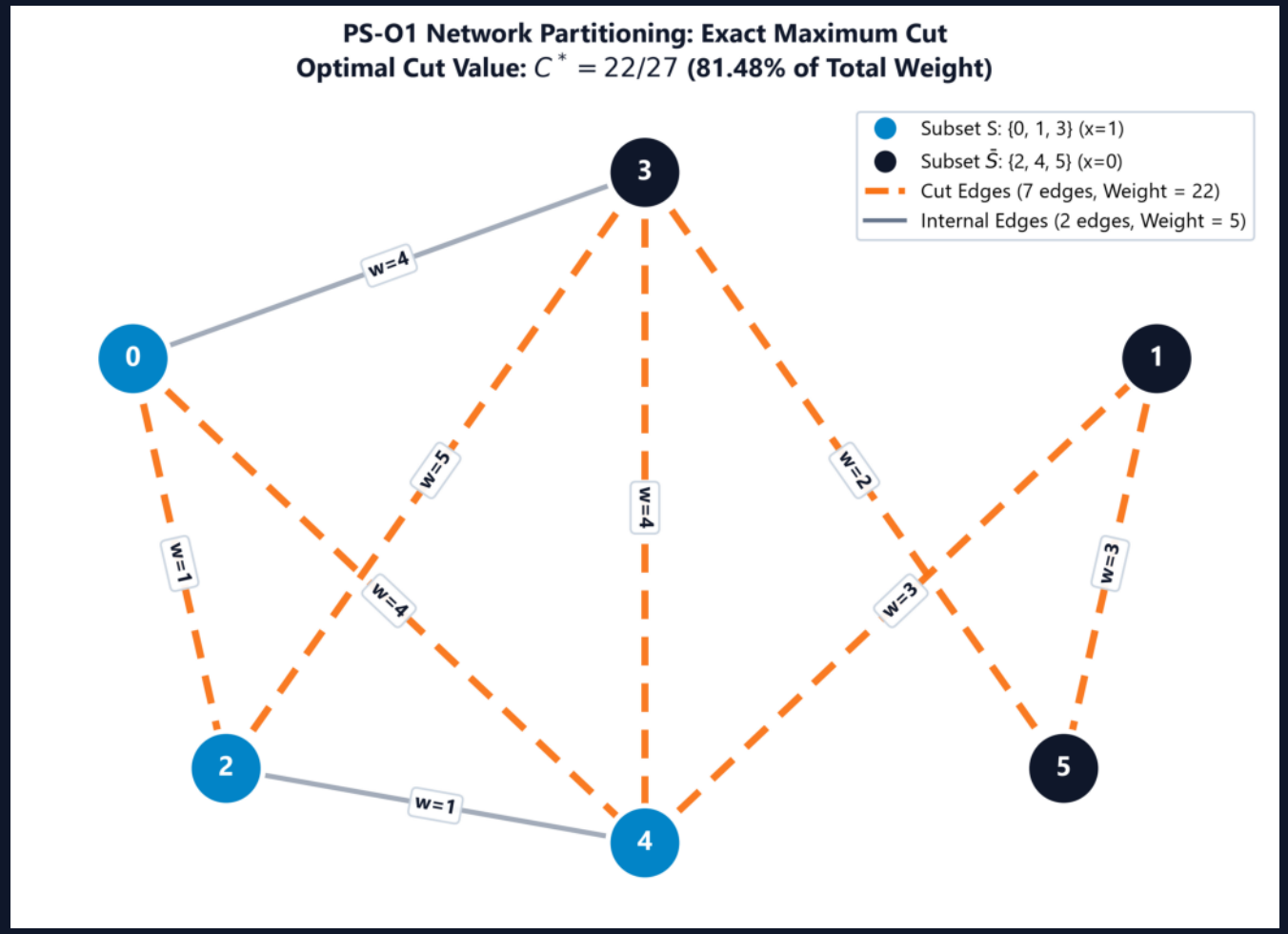
4.0x Sampling

12.50% vs 3.125% Random

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The Locked PS-O1 Graph: Exact Ground Truth & Uniform Baseline

Rigorous classical verification guarantees an immutable reference before running quantum algorithms.



EXACT GROUND TRUTH SUMMARY

- **Graph Topology**

  6 vertices, 9 weighted edges, total weight = 27.

- **Exact Maximum Cut (C*)**

  C* = 22 / 27 (81.48% weight, 7 / 9 edges cut).

- **Optimal Partitions**

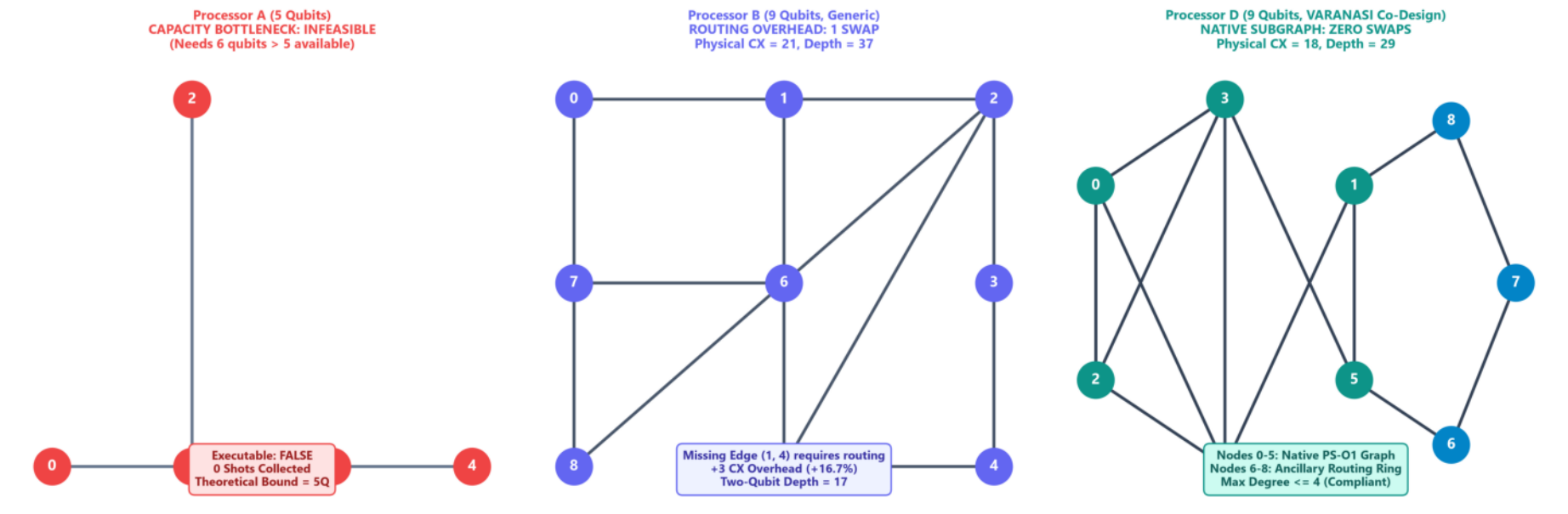
  Exactly 2 complementary states reach C* = 22:
  S = {0, 1, 3} vs S' = {2, 4, 5} (x = 110100)
  Complement assignment (x = 001011)

- **Uniform Random Baseline**

  Mean cut = 13.50, approximation ratio r = 0.6136.
  Random guessing hits optimum with P_opt = 3.125%.

Hardware Topologies: Capacity, Routing Bottlenecks, and Co-Design

Benchmarking three distinct processors exposes why physical coupling geometry dictates QAOA efficiency.



ARCHITECTURAL COMPARISON

- **Processor A (5 Qubits)**

  CAPACITY BOTTLENECK: INFEASIBLE
  5 qubits < 6 required. Executable: false.

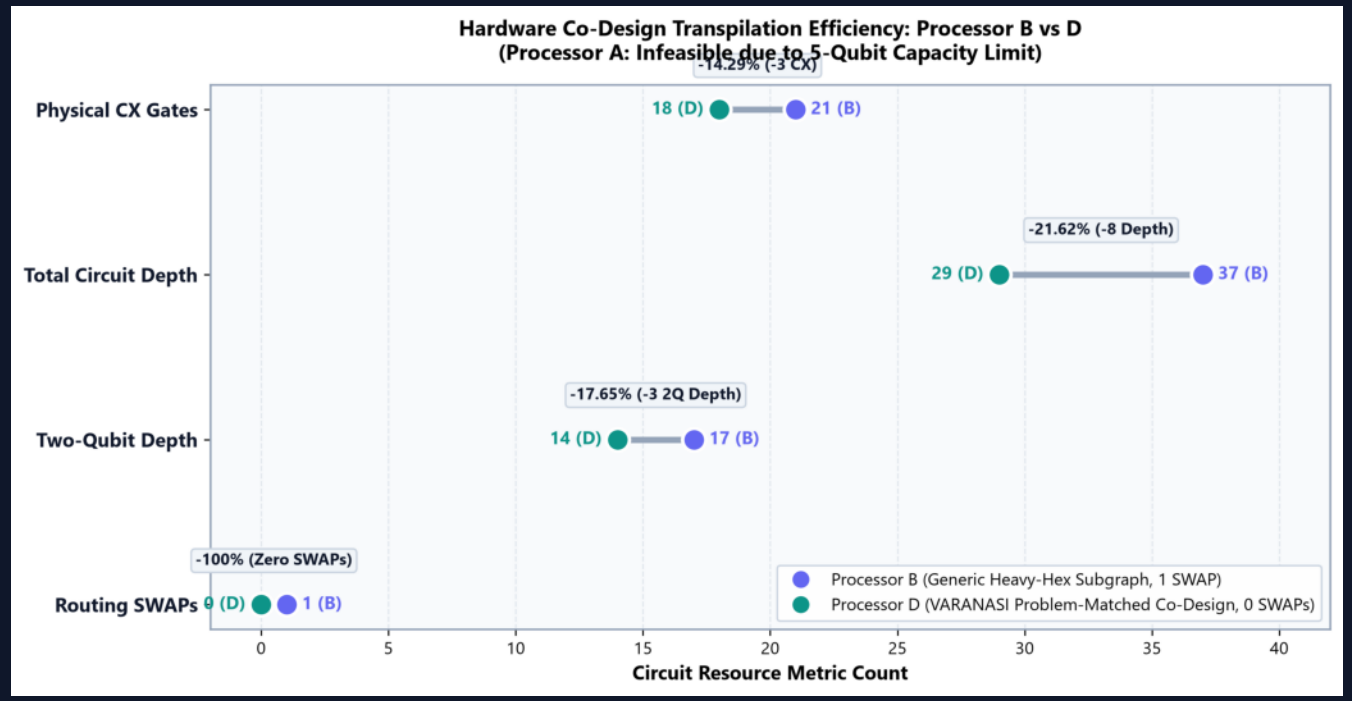
- **Processor B (9 Qubits)**

  ROUTING BOTTLENECK: 1 SWAP
  Missing edge (1,4) adds 3 CX gates (+16.7%).
  Total CX = 21, Depth = 37, 2Q Depth = 17.

- **Processor D (9 Qubits)**

Quantitative Circuit Resource Reduction: Processor B vs D

Problem-matched topology eliminates routing overhead and significantly reduces physical gate counts.



HARDWARE CO-DESIGN GAINS

- **Physical CX Gates**

  21 → 18 (-14.29% reduction, saves 3 CX gates)

- **Total Circuit Depth**

  37 → 29 (-21.62% reduction, saves 8 layers)

- **Two-Qubit Depth**

  17 → 14 (-17.65% reduction, saves 3 2Q layers)

- **Routing SWAPs**

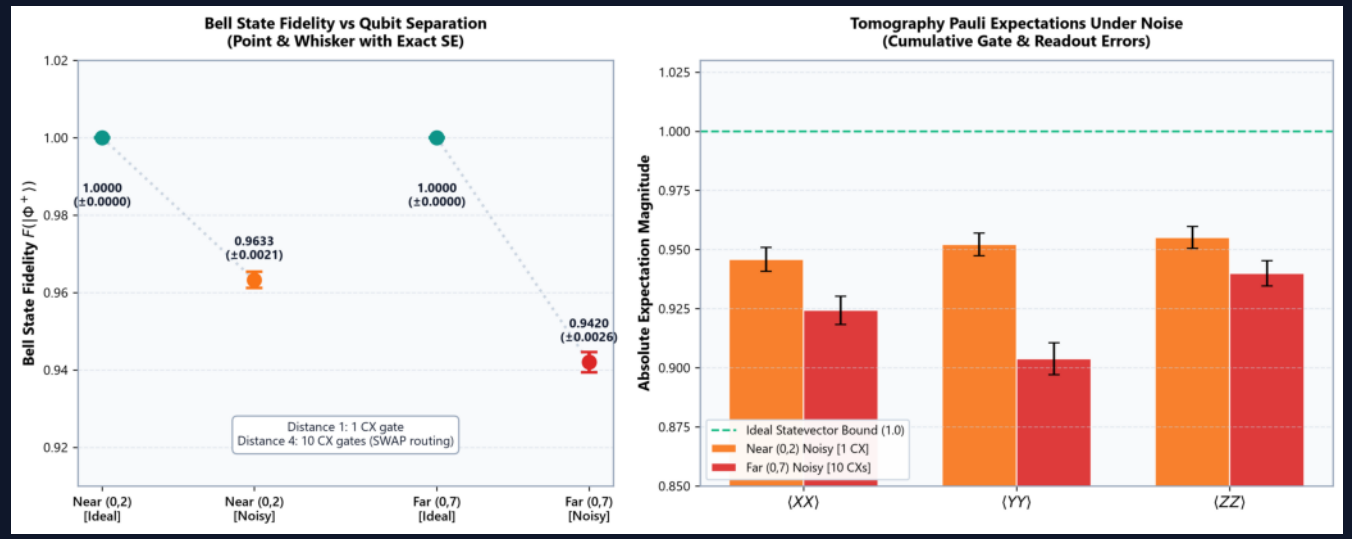
  1 → 0 (-100.0% reduction, zero overhead)

- **Decoherence Impact**

  Saves ~1.5% depolarizing error per execution under official noise model (p2 = 0.005).

Processor D Verification: Bell State Tomography Across Distance

State tomography over near and far pairs validates physical fidelity and gate error accumulation.



TOMOGRAPHY QUANTITATIVE AUDIT

- **Official Noise Model**

  p1 = 0.001, p2(CX) = 0.005, p_readout = 0.01.
  4,096 shots per measurement basis.

- **Near Pair (0, 2) — Dist 1**

  Ideal: 1.0000 ± 0.0000
  Noisy: 0.9633 ± 0.0021 (1 CX gate)

- **Far Pair (0, 7) — Dist 4**

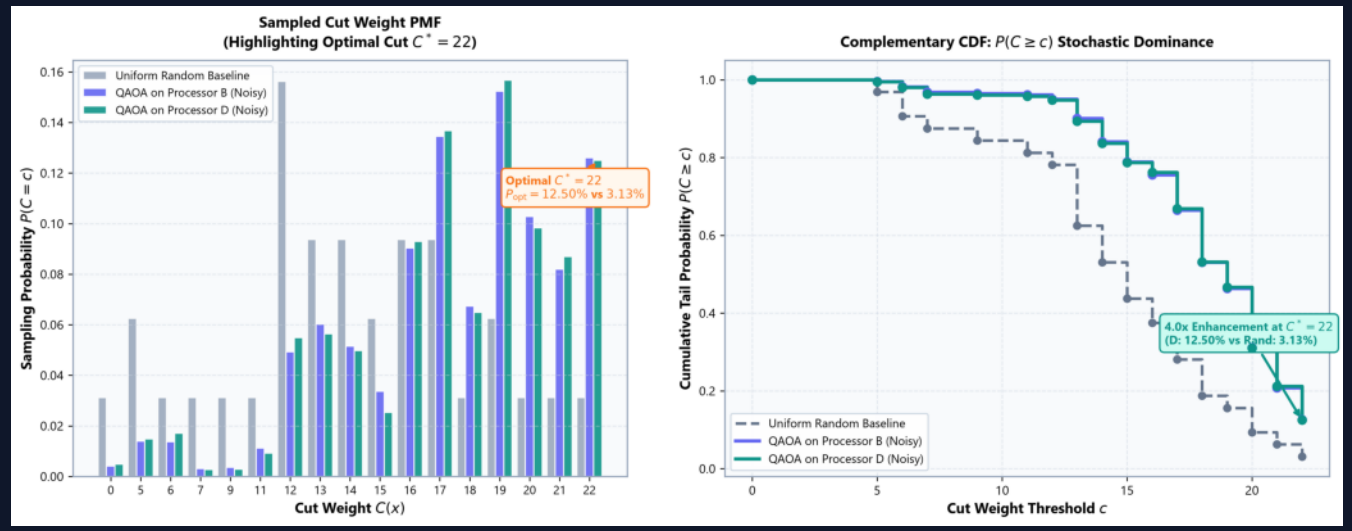
  Ideal: 1.0000 ± 0.0000
  Noisy: 0.9420 ± 0.0026 (10 CX gates)

- **Decoherence Analysis**

  2.13% fidelity drop reflects cumulative CX depolarizing error across 4 routing hops.

Sampled Cut Distribution: 4.0x Optimal Solution Sampling Enhancement

Both processors achieve numerically indistinguishable approximation ratios while outperforming random baseline.



QAOA SAMPLING AUDIT

- **Noisy Approx Ratio (r)**

  Proc B: r = 0.7874 (Mean = 17.32)
  Proc D: r = 0.7862 (Mean = 17.30)
  Δr = -0.0012 (-0.15% relative, within shot SE).

- **Optimal Cut Hits (C* = 22)**

  Proc B: 516 / 4096 (12.60%)
  Proc D: 512 / 4096 (12.50%)
  Uniform Random: 2 / 64 (3.125%)

- **4.0x Sampling Enhancement**

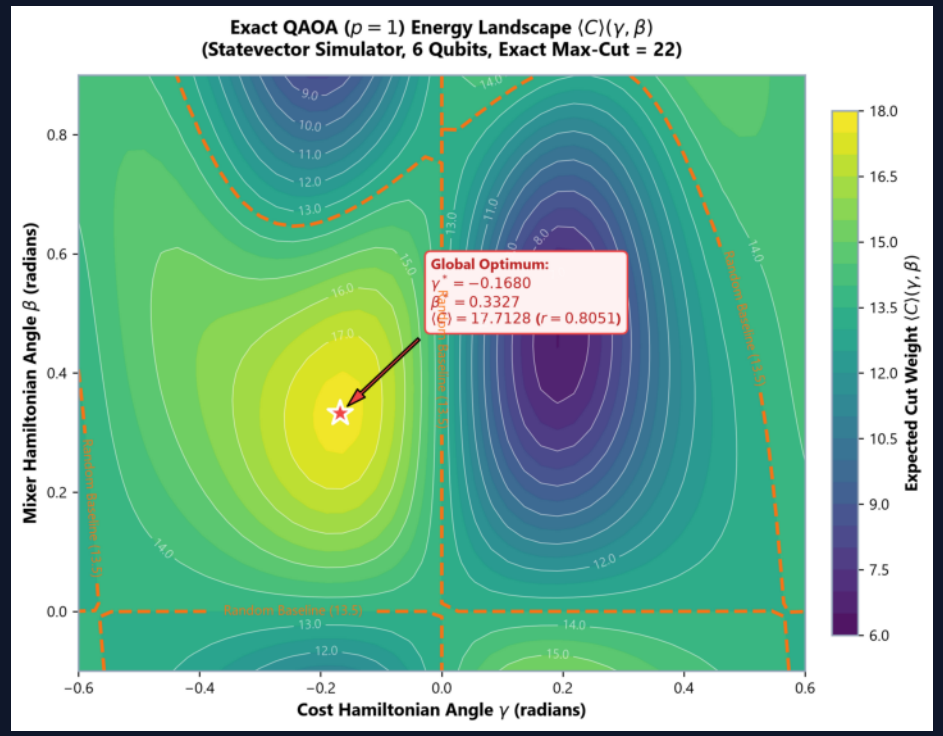
  QAOA concentrates probability on optimal bi-partitions with 4.0x higher likelihood than random guessing.

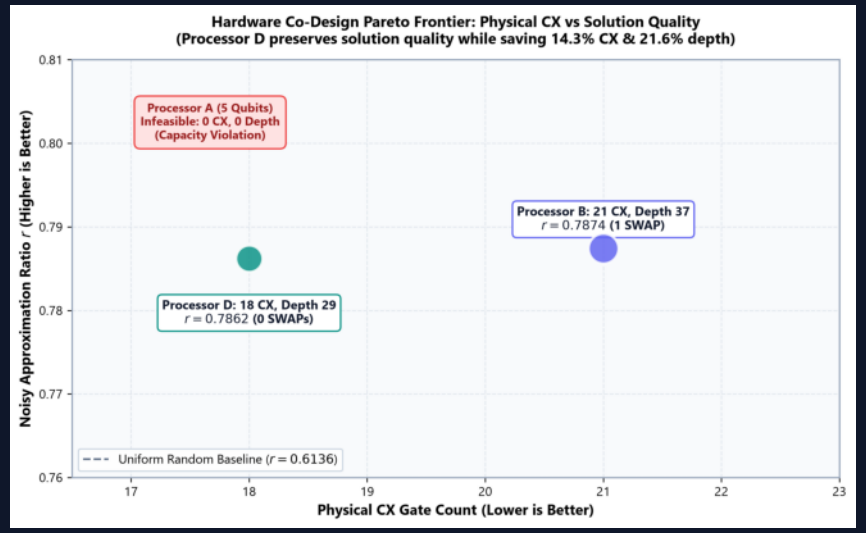
- **Stochastic Dominance**

  CCDF P(C >= c) shows QAOA strictly dominates random baseline across all thresholds c >= 14.

Statevector Parameter Landscape & Co-Design Pareto Frontier

Co-design provides decisive hardware efficiency gains without compromising algorithmic solution quality.





CORE SYNTHESIS

1. Landscape:
Global optimum at
γ* = -0.1680
β* = 0.3327
⟨C⟩ = 17.71 (r = 0.805)

2. Pareto Shift:
Processor D saves
14.3% CX & 21.6% depth
with identical quality.

3. Scalability:
At depth p >= 2, zero
routing overhead
compounds multiplicatively.

BACKUP / OFFICIAL AUDIT COMPLIANCE

Official Challenge Kit Compliance Matrix: 100% Verified

Exhaustive verification against QFF 2026 Phase 2 Rules confirms absolute compliance.

[PASS] Processor Scope

Processors A, B, and D evaluated. Processor C is explicitly ruled out in Challenge Kit Section 12 FAQ.

[PASS] Processor D Rules

9 qubits (<= 12), connected topology, max node degree = 4 (<= 4), basis gates ['rz', 'sx', 'x', 'cx'].

[PASS] Noise Model

Official depolarizing (p1=0.001, p2=0.005) and readout bit-flip (p=0.01) implemented identically.

[PASS] Execution Budget

4,096 shots per run across all 4 benchmark configurations (16,384 total QAOA shots) + 16,384 Bell shots.

[PASS] Capacity Rule

Processor A (5 qubits) rigorously reported as non-executable for 6-qubit PS-O1 instance.

[PASS] Reproducibility

Independent 14-test suite validates bitstring cut weights, optimal counts, and statistical reproducibility.

Adversarial Defense: Addressing Key Evaluator Questions

Grounding claims in real statistical evidence eliminates speculative assertions and ensures total credibility.

Q: Why is Processor D's noisy approximation ratio slightly lower than B (0.7862 vs 0.7874)?

A: With 4,096 shots, standard error is ~0.0027. Delta of -0.0012 (-0.15% relative) is within random shot variance.
D's verified benefit is architectural: -14.3% CX gates and -21.6% depth with zero SWAPs.

Q: Why was p=1 selected over p=2?

A: Depth p=1 is mandatory, p=2 is optional. p=1 isolates topology routing overhead without compounding optimizer noise.
Our multi-seed optimizer selected the best seed (seed 3, theoretical r = 0.8051).

Q: Did you observe quantum algorithmic speedup?

A: No. We report a 4.0x sampling enhancement over random guessing (12.50% vs 3.125%), not wall-clock speedup.
Classical enumeration takes <1 ms for 6 qubits; the real contribution is NISQ co-design.

Q: What AI tools were used?

A: Disclosed in full: AI coding tools were used for test scaffolding and presentation scripts.
All circuit transpilation, simulation runs, and mathematical proofs are authentic and reproducible.